A 195-state PA consistency Turing machine: construction, proof and complete comparison

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14 November 2026 UTC

Abstract

We construct a deterministic binary, one-tape, blank-start Turing machine with **195 working states**, excluding its terminal halt control. The primary construction, called M_{195S}, implements exactly the same 3673-instruction register program as the supplied separate-diagonal-head 196-state machine. A new association of its unchanged 190-token axiom emitter uses 67 reader nodes instead of 68 while retaining six compatible physical reader folds. The complete count is $134 + 67 - 6 = 195$. All 4780 arbitrary-register symbolic source contracts pass, and the literal blank run reaches the exact clean source frame after 58154 transitions.

The proof transfers halting in both directions between the new table and the supplied parent, with positive, terminating source-block simulations. Composing this preservation with the parent's stated mathematical PA calibration gives $\mathsf{PA} \vdash (\mathrm{Nonhalt}(M_{195S}) \leftrightarrow \mathrm{Con}(\mathsf{PA}))$. The archive contains the literal table, reproducible construction, finite certificates, independent audits and the mathematical proof. It does not claim a newly exported end-to-end Lean theorem.

A second 195-state table combines the same decoder with the changed Shift/MP commands and has 3649 source instructions. The comparison inventories all 27 top-level incoming archives and their nested inputs at a fixed revision. It distinguishes the existing successful reports, experimental variants and rejected 195-state files from these new constructions. Earlier bounded failures are retained as research evidence; they are not lower bounds.

Contents

1 Outcome, conventions and evidence **3**

1.1 Fixed source snapshot 3

1.2 What was and was not verified 4

2 How the source designs differ **4**

3 The 195-state construction **5**

4 The 195-state construction and its preservation theorem **6**

5 Secondary command compositions **9**

5.1 Logical emitter inherited from the diagonal construction 9

5.2 Exact command changes 9

5.3 Literal tables and exact blank seed 10

6 Command and source audit for the two new composites **10**

6.1 Exact command interface and emitted stack . . . 10
6.2 Uniform stack-word theorem . . . 11
6.3 Selectors and generalization . . . 12
6.4 Modus ponens, early halting, and a uniform proof compiler . . . 13
6.5 Soundness invariant and scope of the calibration . . . 13
6.6 Finite source binding and arbitrary-value loop proofs . . . 14

7 Reduction experiments and what their failures establish 15
7.1 Parameter names and reassociation . . . 15
7.2 Synthesized helpers and local state deletion . . . 15
7.3 Source shortening and wider macro boundaries . . . 15
7.4 Other inherited ideas tested or assessed . . . 16

8 An exact direct-negation replacement 16

9 The proof endpoint and remaining obligations 18

10 Reproducing the results 18

11 Sources and attribution 19

A Comparison of all supplied PA machine designs 20
A.1 Scope and principal finding . . . 20
A.2 Freshly checked counts . . . 20
A.3 Main lineages . . . 21
A.4 All 27 incoming archives . . . 22
A.5 The 23 designated 196-state graphs . . . 23
A.6 Every explicit 195-state diagnostic . . . 24
A.7 Proof boundaries . . . 25
A.8 Opportunities identified in the supplied designs . . . 25

B Inherited diagonal-calculus calibration 26
B.1 What can be extracted from the new gate . . . 26
B.2 Recovering reverse rewriting . . . 27
B.3 Recovering the neighborhood-equality interface . . . 27
B.4 PA calibration and logical scope . . . 28

C Complete primary transition table 29

1 Outcome, conventions and evidence

The primary deliverable is M_{195S}, a 195-state implementation of the supplied separate-diagonal-head calculus with the original command controller. Its only change is a new binary decoder layout for the *same* axiom-emission word. The full source opcode sequence and every resolved branch target are unchanged. This permits direct preservation of the supplied parent's behavior and consistency calibration.

The numerical saving is one reader control:

$$\underbrace{134}_{\text{fixed controls}} + \underbrace{67}_{\text{new AX readers}} - \underbrace{6}_{\text{compatible folds}} = \boxed{195}.$$

The previous separate-head layout had 68 readers and the same six folds. No change of alphabet, initial input or counting convention is involved.

Construction	Controls	Source sites	Contracts	Blank seed steps
Supplied separate-head parent	196	3673	4780	58154
Primary M_{195S}	**195**	3673	4780	58154
Secondary M_{195C}	195	3649	4749	57922
New diagonal + Combined precursor	196	3676	4784	57922
New separate + Combined precursor	196	3649	4749	57922

The new rows were recomputed here. A contract count is the number of primitive increment, zero/nonzero decrement, jump and halt cases; these are symbolic cases over arbitrary complete register values.

The convention is a two-way-infinite tape, alphabet $\{0, 1\}$, blank symbol 0, head initially at cell 0, and deterministic write–move–next-control transitions. Only left and right moves occur. The primary dense table has working controls 0 through 194, starts in 0, and enters terminal halt 195. It has 390 rows and no row for the halt control. All 195 working controls are graph-reachable from the start.

Let H_e assert arithmetically that the literal blank run of e halts. Writing M_{196S} for the supplied separate-head parent, the preservation result is

$$\mathsf{PA} \vdash (H_{195S} \leftrightarrow H_{196S}).$$

With the parent's stated mathematical calibration, it follows that

$$\boxed{\mathsf{PA} \vdash (\neg H_{195S} \leftrightarrow \mathrm{Con}(\mathsf{PA})).}$$

This is a mathematical proof by preservation of the established design, supported by executable certificates. The distinction from a newly kernel-checked end-to-end theorem is explained in Section9. No finite blank-run prefix is used as proof of nonhalting.

1.1 Fixed source snapshot

All comparisons refer to ProveIt revision

`906367250a7f96c923147b10df392294bced7be5`.

The requested paths were read from that checkout. The imported James Chang PA196 snapshot is `407cdc356f6b586635e71ac92be394255f4bfea4`; the PA209 snapshot is `69d98241a721ead20b4f0b17df9062950b40fbfe`. The archive census includes nested ZIPs by byte identity, not merely top-level filenames. Source paths, table hashes, graph classes and the exact incoming-archive names appear in the inventory appendix and the machine-readable files. The relevant repository sources are listed in Section11.

1.2 What was and was not verified

The released executable checks establish the following finite certificates and symbolic relations for the primary 195-state construction and, where indicated, its variants:

1. exact table reconstruction from the frozen source grammar and six simultaneous reader folds, with no command-row edits for M_{195S};
2. bijective native-to-dense relabelling, all 390 transitions, and graph reachability of all 195 working controls;
3. the complete primitive source/control contracts with symbolic registers and an affine address gap, using the inherited arithmetic-engine block interfaces;
4. exact complete-tape blank initialization to the intended source frame, not merely agreement on the current head and state;
5. free-algebra equality of all generated formula trees and their stack order; and
6. a uniform nine/ten-entry command compiler, including the zero-stack alias and early halting.

The independent literal checker verifies both 195-state tables using a separate implementation, rederiving its generic list lemmas from the exact transitions. It uses no arithmetic-box oracle. Its accelerated symbolic-step counter differs from the compiler's counter; only their semantic conclusions, source cases and literal machine identity are compared.

These are executable certificates with a mathematical preservation argument. They do not constitute a new kernel-checked theorem closing every PA proof-code and arbitrary-model tape-trace interface. In particular, the arithmetic-engine oracle used by the local checker is an explicit inherited interface, not a magically verified black box. The supplied source comments and the command audit identify this dependency.

2 How the source designs differ

The largest apparent improvements in the corpus belong to four different levels. Keeping these levels separate explains why several plausible sub-196 counts fail.

Logical presentation. The source programs enumerate a compact numerical calculus with implication, a binary membership relation, equality and a universal constructor. Replacing a schema, changing the separation gate, or recoding constructors can reduce the emitted formula program. Such a change needs both soundness and PA-strength completeness. It cannot be justified only by a successful numerical simulation of the emitter.

Command program. The finite axiom emitter sits inside an enumerator of command words. Changing the number of emitted stack entries, Shift's halting condition, or where MP stores its result changes the proof-command compiler. Stack selectors and generalization words must be proved again. The new composite audit below treats nine and ten entries uniformly.

Tree representation and reader allocation. A source tree determines binary addresses and structured-loop exits. Reassociation can reduce a shared DAG or improve path length while losing complementary-entry reader folds. A maximum matching in the potential-fold graph can overcount compatible folds: a chosen helper can itself be changed by another fold. Final-table local simulation is therefore required for the entire chosen set.

Physical tape and initialization. The binary runtime distinguishes address zeros from marker zeros, maintains encoded registers and spare blocks, and creates an initial frame from a blank tape. Shortening a scanner or changing spare credit can invalidate this interface even when every high-level arithmetic identity remains true. The runtime-valid U208/PA196 hybrid in the supplied reports is a concrete example: its certified frames simulate the source, but its available blank initializations do not establish such a frame.

The Appendix gives every incoming archive and every designated 196-state graph. It also distinguishes U208's 20 rooted graph classes from its 27 presentations. A different filename or native control numbering is not a new machine; an unrooted graph match alone is not equality of blank behavior.

3 The 195-state construction

The frozen construction is in `construction195/`. Running `build_check.py` without options rebuilds M_{195S}. It reads the supplied parent grammar, expands both grammars to the same 190 macro tokens, resolves both complete primitive programs, and checks exact equality of their opcode/argument sequences and all branch destinations. It then compiles 67 new reader controls, applies the six recorded folds simultaneously, and verifies the final literal machine.

The six pairs below are native pre-fold reader labels. The first control of each pair is removed, its incoming transitions are redirected to the second, and the retained row actions are replaced by the displayed ones. An action (w, D, q) means write w, move in direction D, then enter q.

Removed	Retained	Read 0 action	Read 1 action
2011	2015	$(0, L, 2029)$	$(1, R, 90)$
2018	2051	$(0, R, 166)$	$(1, L, 2007)$
2020	2061	$(0, L, 2059)$	$(1, R, 90)$
2024	2033	$(0, L, 2009)$	$(1, R, 61)$
2026	2048	$(0, L, 2034)$	$(1, R, 2029)$
2054	2055	$(0, L, 2017)$	$(1, R, 2039)$

These are history-sensitive reader folds: the preceding tape bit distinguishes the two entry contexts. The local checker verifies all four predecessor/read-bit cases for each pair in the *final* folded table, preserving the complete local tape and the appropriate successor/head position. Thus helper interactions are checked after all six changes, not only in six isolated parent tables.

The default construction preserves all 134 fixed controls and every arithmetic-box transition. Its final table has 195 controls and 390 rows. The exact source program has 3673 sites and 4780 primitive cases. The universal symbolic runtime audit passes all cases, with maximum 452 accelerated transition steps in one contract. Literal initialization reaches the full clean frame after 58154 transitions, at native 84, head position-2, source site 25, address-gap parameter 48, $N = 2$ and the other logical registers zero, with 26 spare blocks of value 1. The full tape support is compared.

The optional `--combined` construction M_{195C} uses exactly the same 67-node grammar and six folds, together with the already audited command composition. It has 3649 source sites, passes 4749 symbolic cases, and initializes in 57922 transitions. It is supplied as an additional construction; the direct-preservation version M_{195S} is the primary result.

The decoder was discovered after exact fold compatibility was moved inside the grammar search's fitness evaluation. The successful grammar uses the identity permutation of all head parameters and the original head order. Thus neither a renamed logical schema nor changed selector constants are needed for the primary construction. A separate associativity witness records

371 occurrence-level rotations from the parent tree through a canonical association. This is a word-equality witness, not a claim that every intermediate tree is a valid physical machine. Endpoint control-flow equality is checked directly.

Independent universal physical verification. For each of the exact literal tables 195S and 195C, the independent affine whole-tape checker verifies every primitive source outcome: respectively 4,780 outcomes of 3,673 sites and 4,749 outcomes of 3,649 sites. Each contract ranges over arbitrary natural values of all nine primary registers, arbitrary nonempty finite spare lists, and every initial header gap at least 48. Both decrement outcomes are checked without a source-reachability restriction. Successful checking requires equality of the complete symbolic tape, the prescribed control and head displacement, and strictly positive elapsed time.

This checker uses literal transition rows, elementary induction for unary self-loop sweeps, and four generic finite-list lemmas rederived from each new table with arbitrary opaque surrounding tape. It does not replace arithmetic routines by register-box oracles. The per-outcome affine step counts and motions, the one-block lemma traces, and the exact finite blank seeds are retained in the certificate. The complete blank runs reach the required source-site-25 seed after exactly 58,154 and 57,922 steps. Dense/native conjugacy and both independent source CFG resolutions are checked as well. This verifies the physical source-to-tape interface; the logical calibration and the end-to-end internal arithmetic statement remain separate mathematical proof obligations.

4 The 195-state construction and its preservation theorem

Let M_{196}^{sep} be the separated-diagonal machine supplied in `pa_consistency_comparison_and_new196_2026-11-13.zip`, with the original PA196 command control. Let M_{195} be the new table in this archive using the same fixed control and the new axiom-reader grammar. Both machines use a two-way infinite binary tape, initially all zero. In native labels the initial state is 74 and the unique halting label is 254. A dense relabeling does not affect the theorem.

What the construction changes. The new grammar has 67 distinct reader nodes. The fixed part has 134 working states, and six disjoint, simultaneously valid context folds remove six reader states. Hence the number of working states is

$$134 + 67 - 6 = 195.$$

There are exactly 390 transition rows, one for each working-state and symbol pair. Every working state is graph-reachable from the initial state. The halting label is not counted as a working state.

The flattened axiom-emitter word has exactly the same 190 tokens, in the same order, as the supplied separated-diagonal emitter. Although the search allowed independent parameter permutations, the final grammar uses the identity permutation in all six heads. No axiom formula, parameter assignment, guard, stack position, or command selector changes. The new object is a different binary parenthesization and sharing of the same emitter. All fixed transition rows are unchanged except the reader entry transition $(101, 1)$, which is redirected to the new reader root.

Token equality alone would be insufficient for this tree language: changing ancestors can change the targets of structured jumps. The stronger check is exact equality of the entire branch-resolved register program. Both source trees have 3,673 primitive sites. Number their leaves from left to right by $0, \ldots, 3672$. At every ordinal, the instruction, register operand, and every successor ordinal agree. This includes both branches of every decrement, every break and continue target, and every halt. Only source addresses in the binary tree change. The equality is checked using both arithmetic path cutting and an independent resolver based on subtree intervals and ancestor

walking. The latter also checks that a proposed successor has no nonzero suffix beyond its target leaf.

Source identity lemma. Let G be this common finite control graph, acting on nine natural-number registers. The two source programs have identical executions after identifying their program counters by leaf ordinal. This holds for every initial register vector, not merely the clean seed. Indeed, the step functions are literally the same finite case distinction: increment has the same register and successor; decrement tests and updates the same register and has the same two successors; a jump has the same target; and halt has the same meaning. Induction on the number of source steps proves equality of their finite traces. Since the graph is finite and explicit, PA proves the same statement about its coded step functions.

The inherited tape interfaces. The source-to-tape argument uses the original PA196 interfaces for scanning and marking a unary register, incrementing it, and decrementing it with its zero/positive case distinction. These interfaces quantify over the complete arbitrary register file and the intervening tape gap. They specify the action on that whole file: in particular, increment also appends a one-valued spare register, as in the inherited engine. The untouched logical registers are preserved, and the file continues to satisfy its minimum-length invariant. The interfaces also assert termination. The register routines in M_{195} are literally unchanged. More precisely, the new table agrees on all 44 entries of `M6.Native.boxRows` and on the append-one tail $(198, 0) \mapsto (1, L, 40)$. The parameterized theorem `M6.Native.boxLemmas` therefore supplies the same four register contracts for this table. In the upstream formal development these are the block contracts supplied to the symbolic executor in `M6/Runtime.lean`, whose soundness chain includes `oracle_sound`, `sstep_sound`, and `exec_sound`. The present executable audit uses those mathematical block interfaces; it does not replace them by bounded tests on numerical register samples.

For each primitive source site, and for both branches when appropriate, the delivered symbolic certificate begins at the full encoded source frame. It reaches exactly the next full frame with the specified register effect, or reaches the halting state for a halt instruction. The symbolic gap is affine in an arbitrary natural number, and the register file is an arbitrary symbolic object satisfying the stated branch condition. The 195-state certificate has 4,780 cases, a maximum of 452 symbolic steps in a case, and 1,442,472 symbolic steps in total. These are symbolic proof-case counts, not bounds on the physical execution time for arbitrarily large register values.

Every nonhalting source instruction is simulated by a finite, positive number of physical steps, without a premature halt. The positive-length condition is explicit in the executor: reaching the target does not close a case until an actual tape transition has occurred. Compressed constant-symbol scans and register oracles represent finite runs whose lengths have elementary, hence primitive recursive, bounds in the frame parameters. This excludes an infinite physical computation trapped inside one source instruction. The complete source graph audit supplies the correct target for each symbolic case.

The six reader folds have an additional local check. Before folding, each member of a pair is entered by a rightward, symbol-preserving move, with the bit just crossed distinguishing the two members. One member has only incoming context zero, and the other only incoming context one. For each of these two contexts and each current symbol, the replacement follows exactly the original transition's tape effect, head displacement and target, allowing a finite detour that inspects the previous bit. All 24 cases are checked against the final table after all six folds. The detours stay within the two cells described by the local interface, so the arbitrary tape outside them is untouched. The whole-frame symbolic certificate then checks the folded table itself, not only its unfolded parent.

Blank initialization. Both the supplied parent and the new table reach their exact clean source frames after 58,154 literal tape transitions from the all-zero tape. The source counter is site 25 and the logical register vector is $(2, 0, 0, 0, 0, 0, 0, 0, 0)$. The complete tape, head position and native dispatch state 84 are checked, including the unchanged dummy-register suffix. This is a fixed finite trace, so its correctness is an elementary PA calculation. Agreement only on a selected portion of the tape would not suffice here; the delivered check compares the complete finite support of the tape.

Finite-trace equivalence theorem. Write H_M for the arithmetic sentence that M has a finite halting trace from its specified blank start. Then

$$\mathrm{PA} \vdash H_{M_{195}} \iff H_{M_{196}^{\mathrm{sep}}}. \qquad \text{(195-preservation)}$$

To prove this inside PA, use the common source graph G and the exact clean seed. A finite halting source trace produces a finite halting tape trace by concatenating the exact startup and primitive execution segments. Their codes and finite bounds are primitive recursively constructible, and PA proves their correctness by induction on the length of the source trace.

For the reverse implication from a tape trace to a source trace, suppose the tape trace first halts after T transitions. Iterate the common source step function for $T+1$ instructions from the clean seed. If none of these instructions halts, their exact positive-length simulations, after startup, concatenate to a physical run longer than T, with no halt before its end. Determinism contradicts the alleged first halting trace. Thus one of the first $T+1$ source instructions halts. This bounded iteration is primitive recursive; its correctness needs only ordinary arithmetic induction. Apply both directions to each of the two machines and eliminate the common source-halting predicate. The argument does not assume that either machine actually fails to halt.

Composition with the PA calibration. The numerical calculus and proof-search program in this theorem are exactly the supplied separated-diagonal ones. The inherited mathematical calibration gives

$$\mathrm{PA} \vdash H_{M_{196}^{\mathrm{sep}}} \iff \exists p\ \mathrm{Prf}_{\mathrm{PA}}(p, \ulcorner 0 = 1 \urcorner). \qquad \text{(parent-calibration)}$$

Its substantive interfaces are the following. The arithmetic interpretation translates membership to finite binary-set membership and supplies PA proofs of pointed separation by full induction. The formula-code freshness lemma supplies proofs of the diagonal clause $B \to U(B, B)$. The recovered two-direction membership rewrites, neighborhood equality and equality-witness interfaces feed the original PA-refutation compiler. The separated diagonal head uses the adjusted ten-item stack selectors. In the soundness direction, a proof annotation of the top, cached head and canonical Cantor stack turns a source halt into a PA refutation, including the zero-stack alias. In the completeness direction, the proof-command compiler and exhaustive command scheduler turn a PA refutation into a source halt. These are primitive recursive syntactic constructions with PA-provable correctness; no uniform truth predicate or reflection principle for PA is used. The inherited calibration and diagonal-interface arguments are included elsewhere in the report so that the logical dependency is visible.

Combining parent calibration with (195-preservation) and the definition $\mathrm{Con(PA)} \equiv \neg\exists p\ \mathrm{Prf}_{\mathrm{PA}}(p, \ulcorner 0 = 1 \urcorner)$ gives the requested result:

$$\boxed{\mathrm{PA} \vdash (M_{195} \text{ does not halt from its blank start}) \iff \mathrm{Con(PA)}.}$$

This is an ordinary mathematical proof obtained by preservation and composition with the stated upstream interfaces. The executable files make the finite construction and its certificate data

reproducible. They do not constitute a newly exported Lean theorem for the entire displayed biconditional. The inherited formal development and the additional written arithmetic-calibration argument are distinct parts of the proof provenance.

5 Secondary command compositions

5.1 Logical emitter inherited from the diagonal construction

Let

$$p(a,b) = \binom{a+b+1}{2} + b, \quad I(a,b) = 4p(a,b), \quad U(a,B) = 4p(B,a) + 2, \quad \neg B = I(B,1).$$

Code 0 is the special truth alias $I(0,0) = 0$; code 1 is false. The diagonal construction uses

$$\Gamma(B) = I(B, U(B,B)), \qquad B \longrightarrow \forall x_B B.$$

Every variable label in the decoded nonzero formula B is strictly less than B, by induction on the numerical constructors. Hence x_B is fresh for B. Code 0 has no free labels. This is the reason the diagonal family is sound; arbitrary universal generalization under an assumption is not being asserted.

Write V and Q for the unchanged membership-vacuity and pointed separation expressions of the supplied calculus. The diagonal emitter uses the guarded bundle

$$F_\Gamma(a,b,c) = \neg(\Gamma(c) \to (V(a,b,c) \to Q(a,b,c))).$$

The guard $c \neq a, b$ remains. The separate-head variant leaves $\Gamma(c)$ as an additional unconditional stack item and uses the old guarded gate

$$G(a,b,c) = \neg(V(a,b,c) \to Q(a,b,c)).$$

The two presentations have fixed classical proof translations: F_Γ gives both Γ and G, while Γ, G give F_Γ. To obtain any diagonal instance $\Gamma(B)$ from the bundled version, choose $c = B$ and $a = b = B + 1$; the guard succeeds. Thus no unrestricted substitution into a freshness-restricted schema is assumed.

The supplied diagonal completeness argument recovers reverse membership rewriting and the neighborhood-equality interface by choosing the fresh label to be the code of the negated entire conclusion. It retains membership vacuity. Merely deleting that vacuity, or deleting the first quantifier head without a replacement for explosion, is not justified by the same argument. The full inherited diagonal derivation and its source provenance are included under `logical_background/`; the new work here is its composition with the changed commands and the validation of the resulting literal tables.

5.2 Exact command changes

The new tables use the supplied Combined196 implementation. In native control names its six row edits, relative to the pinned published PA196 fixed part, are:

Control	Read	Write	Move	Next
38	0	0	R	11
38	1	1	R	122
164	1	1	R	130
214	0	0	R	312
214	1	1	R	20
312	0	0	R	188

These names are local to the pinned PA196 conjugacy. In particular, they must not be copied into a U208 or ZFC table with equal-looking integer labels. The source changes replace the successful MP tail by a transfer into the cached head and use a Shift that tests that head alone. The source-binding checker identifies the exact retargeted exits and changed tail.

For either axiom grammar the initial compilation has 202 controls: 134 unchanged fixed controls and 68 emitted reader controls. Six explicit simultaneous folds give

$$134 + 68 - 6 = 196.$$

The fixed part includes the arithmetic engine, leaf routing and the non-AX readers. Each fold certificate is checked in the final folded table, with tape, head and successor agreement. Merely adding the individual savings from the parent reports would not justify this sum.

5.3 Literal tables and exact blank seed

The complete tables are `axioms/diagonal196/machine.tm` and `axioms/separate196/machine.tm`. Their sparse native versions, native/dense bijections, grammar DAGs, source paths, fold certificates and receipts are adjacent. Native initial control 74 corresponds to dense 0, and native terminal 254 corresponds to dense 196.

Both new machines reach exactly the expected complete frame after 57922 blank-tape transitions. At that frame the native control is 84, the head is at cell -2, the source site is 25, the address gap parameter is 48, logical register N is 2, the other eight logical registers are zero, and there are 26 spare blocks of value 1. The physical register permutation is

$$(N, T, W, A, D, H, X, Y, L) \longmapsto (6, 2, 0, 4, 3, 7, 8, 5, 1).$$

The checker compares the entire finite support of ones, including both markers and all register/spare encodings. It neither stops at a merely plausible control nor accepts an approximate tape match.

The symbolic runtime checks 4784 and 4749 primitive cases, respectively. The largest accelerated microtrace has 497 counted transition steps in the compiler's executor. The full symbolic state represents arbitrary complete register values and a symbolic gap; this is stronger than testing a finite grid of small inputs. It is still conditional on the declared arithmetic-engine contracts. The independent implementation checks the same source coverage and refines the reachable predecessor contexts used in the reduction search.

6 Command and source audit for the two new composites

The two composites combine the supplied diagonal-vacuity axiom emitters with the supplied changes that make Shift test the cached head alone and make successful modus ponens return its result in that cache. These changes are compatible. The ten-entry emitter requires changed stack indices; the nine-entry command compiler cannot simply be copied verbatim. This section proves the required command identities uniformly, binds the changed primitive source tails, and states the limits of the audit.

6.1 Exact command interface and emitted stack

Let $p(a, b) = \binom{a+b+1}{2} + b$, let $[a; w] = p(a, w)$, and write $[a_1, \ldots, a_k; w]$ for iterated cons with a_1 at the top. The total Cantor inverse satisfies

$$\mathrm{hd}[a; w] = a, \qquad \mathrm{tl}[a; w] = w, \qquad \mathrm{hd}(0) = \mathrm{tl}(0) = 0.$$

In particular, $[0;0] = 0$. No argument below assumes that a zero push increases a stored stack length, or that an empty stack can be distinguished from a stack of zero entries. The identities are numerical identities for the total projections, valid also at this alias.

Use $I(a,b) = 4p(a,b)$ and $U(a,B) = 4p(B,a)+2$, and write a command store as $\langle t,w,d,h\rangle$. The two modified commands are

$$S\langle t,w,d,h\rangle = \begin{cases} \mathsf{halt}, & h = 1, \\ \langle h, \mathrm{tl}(w), d, \mathrm{hd}(w)\rangle, & h \neq 1, \end{cases}$$

and, putting $y = d + x$,

$$\mathsf{MP}_x\langle t,w,d,h\rangle = \begin{cases} \langle 0,w,0,y\rangle, & I(t,y) = h, \\ \langle 0,w,y,0\rangle, & I(t,y) \neq h. \end{cases}$$

Only S can halt. Neither branch of MP pushes a stack entry.

Let n be the *total* number of entries AX pushes, including the saved top: $n = 9$ for the diagonal composite and $n = 10$ for the separate diagonal-head composite. For $D = d + x$, $H = h + y$, $A = c$, the effect is

$$\mathsf{AX}_{c,x,y}\langle t,w,d,h\rangle = \langle K(H,D,c), [q_1,\ldots,q_{n-1},t;w], 0, 0\rangle.$$

Writing $L_1,\ldots,L_6$ for the original six heads in their push order, the generated entries, listed from the stack top downward, are exactly

n	$(q_1,\ldots,q_{n-1})$
9	$(L_6,L_5,L_4,L_3,L_2,L_1,\rho_8(D,A,H,d),U(d,h))$
10	$(\Gamma(A),L_6,L_5,L_4,L_3,L_2,L_1,\rho_8(D,A,H,d),U(d,h))$.

Here $\Gamma(B) = I(B,U(B,B))$. In the first case K is the guarded bundle $F_\Gamma = \neg(\Gamma(c) \to (V \to Q))$; in the second it is the guarded old gate $G = \neg(V \to Q)$. A failed numerical guard gives $K = 0$, while the separately pushed $\Gamma(A)$ remains present in the second case. The emitter audit checks these exact free-algebra trees and the complete stack order, not a list of sampled formula codes.

All q_j except the last have residue zero modulo four; the last is $U(d,h)$ and has residue two. The gate, including a failed guard, has residue zero. Consequently none of these generated values is code one. This property is enough for all the termination claims below; it does not assume that a theorem of the underlying calculus is numerically nonzero.

6.2 Uniform stack-word theorem

Commands execute from left to right. For every incoming store, including one with $t = 1$, the following are total, nonhalting command executions:

$$\mathsf{AX}_{c,x,y}; S^{n-1} : \langle t,w,d,h\rangle \longmapsto \langle \rho_8(d+x,c,h+y,d), [t;w], 0, U(d,h)\rangle,$$
$$\mathsf{AX}_{c,x,y}; S^{n} : \langle t,w,d,h\rangle \longmapsto \langle U(d,h), w, 0, t\rangle.$$

Indeed, the first Shift tests the cleared cache zero; Shifts two through n test q_1 through q_{n-1}. None tests the saved top t. This is a finite induction using the two Cantor projection identities, so it also covers $t = w = 0$ and all other zero aliases.

Define

$$\mathsf{Push}_n = \mathsf{AX}_{0,0,0}; S^{n-1}; \mathsf{MP}_0,$$
$$\mathsf{Pop}_n = \mathsf{AX}_{0,0,0}; S^{n+1},$$
$$\mathsf{Discard}_n = \mathsf{MP}_1; \mathsf{AX}_{1,0,0}; S^{n}; \mathsf{MP}_0.$$

Their complete contracts are

$$\begin{aligned}
\mathsf{Push}_n\langle t, w, d, h\rangle &= \langle 0, [t; w], 0, 0\rangle,\\
\mathsf{Pop}_n\langle t, [a; w], d, h\rangle &= \begin{cases} \mathsf{halt}, & t = 1,\\ \langle t, w, 0, a\rangle, & t \neq 1,\end{cases}\\
\mathsf{Discard}_n\langle t, w, 0, 0\rangle &= \langle 0, w, 0, 0\rangle.
\end{aligned}$$

Push's last MP compares an implication, of residue zero, with $U(d, h)$, of residue two, and therefore takes its failure branch. Pop's last Shift tests precisely the saved top; this is the only possible halt in that word. For Discard, $I(t, 1) > 0$ makes the first MP fail, producing $\langle 0, w, 1, 0\rangle$. The AX identity gives $\langle U(1, 0), w, 0, 0\rangle$, and $I(U(1, 0), 0) > 0$ makes the final MP fail. These arguments cover arbitrary t, including zero and one.

Thus the concrete exponents are

	$\mathsf{Push} : S^{n-1}$	$\mathsf{Pop} : S^{n+1}$	$\mathsf{Discard} : S^n$
diagonal	8	10	9
separate diagonal head	9	11	10.

In particular, Pop is *not* claimed total for arbitrary preserved top: it deliberately detects a derived contradiction.

6.3 Selectors and generalization

Let Offset_0 be empty and let $\mathsf{Offset}_u = \mathsf{MP}_u$ for $u > 0$. Then

$$\mathsf{Offset}_u\langle 0, w, 0, 0\rangle = \langle 0, w, u, 0\rangle,$$

since $I(0, u) > 0$ when $u > 0$. Define

$$\mathsf{Gen}_{i,n} = \mathsf{Push}_n; \mathsf{Offset}_i; S; \mathsf{AX}_{i,0,0}; S^n.$$

For every a, w, including $a = 1$,

$$\mathsf{Gen}_{i,n}\langle a, w, 0, 0\rangle = \langle U(i, a), w, 0, 0\rangle.$$

The Shift before AX tests cache zero and retrieves a into h; AX clears that cache before any subsequent Shift and constructs $U(i, a)$. The last identity from the preceding subsection then proves the result.

For $1 \leq j \leq n - 3$, the ordinary head selector is

$$\mathsf{Discard}_n; \mathsf{AX}_{A,D,H}; S^{j+1}; \mathsf{Pop}_n^{n-1-j}.$$

It returns $\langle q_j, w, 0, 0\rangle$ from $\langle t, w, 0, 0\rangle$ for arbitrary t, w. After the displayed Shifts, $t = q_j$, the cache contains q_{j+1}, and the stack is $[q_{j+2}, \ldots, q_{n-1}, 0; w]$. There are exactly $n - 1 - j$ entries in that displayed stack segment. Each Pop preserves $q_j \neq 1$, consumes one entry and clears the previous cache before its own Shift tests. The last consumed entry is the saved zero, giving cache zero.

For $n = 10$, the new diagonal head is $j = 1$, so the concrete selector for $\Gamma(B)$ is

$$\mathsf{Discard}_{10}; \mathsf{AX}_{B,0,0}; S^2; \mathsf{Pop}_{10}^8.$$

This is valid even for a failed gate guard and even for $B = 0$. Each original head has index one larger than in the nine-entry layout.

For $u \le a$, the restricted propositional head is selected by

$$\mathsf{Discard}_n; \mathsf{Offset}_u; \mathsf{AX}_{b,a-u,c}; S^{n-1}; \mathsf{Pop}_n,$$

which returns $\rho_8(a,b,c,u)$ with the original stack and zero caches. The gate selector is

$$\mathsf{Discard}_n; \mathsf{AX}_{c,b,a}; \mathsf{Pop}_n^n.$$

It returns $K(a,b,c)$ with the original stack and zero caches. Every Pop is safe because its preserved top is the gate, which has residue zero. These selectors are total even when their incoming top is one.

6.4 Modus ponens, early halting, and a uniform proof compiler

The MP suffix is independent of the AX stack length:

$$\mathsf{MPSuffix}_b = S; S; \mathsf{MP}_b; S.$$

For every a, b, w,

$$\mathsf{MPSuffix}_b\langle 0, [a, I(a,b), 0; w], 0, 0\rangle = \begin{cases} \mathsf{halt}, & a = 1 \text{ or } b = 1, \\ \langle b, w, 0, 0\rangle, & a \neq 1 \text{ and } b \neq 1. \end{cases}$$

The first Shift retrieves a. The second either halts at $a = 1$ or sets the top to a and the cache to $I(a,b)$. MP succeeds and puts b in the cache. The last Shift either halts at $b = 1$ or retrieves the dummy zero, leaving the underlying stack unchanged. If $a = b = w = 0$, all three displayed entries alias the zero stack. The same projection calculation still applies and MP succeeds at $I(0,0) = 0$. Omitting the dummy zero would consume the first underlying entry, so its presence is essential.

For a finite proof tree p in the appropriate numerical calculus, compile axioms by the selectors above and compile generalization by $W_{p_0}; \mathsf{Gen}_{i,n}$. For an MP node with minor premise a and major premise $I(a,b)$, use

$$W_p = \mathsf{Discard}_n; \mathsf{Push}_n; W_{p_1}; \mathsf{Push}_n; W_{p_0}; \mathsf{Push}_n; \mathsf{MPSuffix}_b.$$

Induction on p proves the exact invariant: for every t, w, starting from $\langle t, w, 0, 0\rangle$, the compiled word either halts or returns $\langle \mathrm{concl}(p), w, 0, 0\rangle$. The first two words in the MP case introduce the dummy zero; the two premise computations and their Pushes then establish the suffix's exact three-entry precondition. Early halting is preserved by command-word extension and causes no completeness failure.

If p proves code one, the word

$$\mathsf{AX}_{0,0,0}; W_p; \mathsf{Pop}_n$$

halts from *every* incoming store. The first AX clears d, h; the compilation invariant tolerates its arbitrary resulting top and stack. If no earlier command halts, the final Pop detects the proven top one before inspecting the arbitrary underlying stack. This universal incoming store property is exactly what is needed for the inherited exhaustive command-word scheduler; earlier trials need not restore an empty store.

6.5 Soundness invariant and scope of the calibration

Maintain derivations in the chosen numerical calculus for t, h, and each finitely decoded stack entry. No derivability requirement is placed on the offset d. Code zero has the fixed derivation

$\rho_8(0,0,0,0) = 0$, because $I(0,0) = 0$. Thus the infinite zero tail does not require infinitely much proof data: a single fixed zero proof supplies every requested padding entry.

AX preserves this invariant: its saved top is already derived, its ordinary heads and guarded gate are axioms or zero, the separate $\Gamma(A)$ is an unconditional axiom, its propositional head satisfies $d \le d + x$, and $U(d,h)$ follows by theorem generalization from the old cache. A failed MP returns zero; a successful MP derives $d + x$ from the derived t and $I(t, d + x) = h$. Shift only moves derived formulas, and a halting Shift exposes a derivation of one. Together with the compiler and the inherited scheduler, this proves command-program halting if and only if the selected numerical calculus has a refutation.

The compiler, selectors, finite proof-store updates, and the two changed loop-trace constructors are primitive recursive. The correctness arguments use ordinary induction on finite proofs, lists, natural loop counters and traces. They therefore supply the changed-command part of a PA-internal calibration. This audit does not independently reprove the entire inherited PA-to-calculus and calculus-to-PA translation, nor claim a newly checked Lean or Rocq endpoint. Those logical translations and the internal coding of the literal tape simulation remain the inherited interfaces required for an end-to-end assertion about Con(PA).

6.6 Finite source binding and arbitrary-value loop proofs

An independent pair of CFG resolvers checks the complete source graphs. Let N be the new program's number of primitive sites. Set $E = N - 15$, $F = N - 12$, $S_0 = N - 11$. Its first E primitive instructions are identical to those of the corresponding old axiom layout, except for three exit retargetings: source site 48 branches to the new Shift entry, the last AX site $N - 96$ returns to F, and the last MP-prelude site $N - 16$ returns to F. No AX macro or formula-emission instruction changes. The concrete entries are

	N	AX final	MP success E	Shift S_0
diagonal composite	3676	3580	3661	3665
separate composite	3649	3553	3634	3638

The new MP tail is the three-instruction loop $\operatorname{dec} D; \operatorname{inc} H;$ jump followed by return to fetch site 25. From $(D, H) = (y, 0)$ it reaches $(0, y)$ in exactly $3y + 1$ instructions and returns in one more. Failure returns directly with $(D, H) = (y, 0)$. This proves the displayed MP contract for every natural y.

The inherited command-boundary convention gives scratch $X = 0$; the four stored values t, w, d, h remain arbitrary. The new Shift first clears T in $2t + 1$ instructions, then transfers H to T and scratch X in $4h + 1$ instructions, then restores H from X in $3h + 1$ instructions. Thus it reaches $(T, H, X) = (h, h, 0)$ after exactly $2t + 7h + 3$ instructions. Its final two instructions set $X = 1$ and enter the unchanged guard, which compares H to one and otherwise calls the unchanged total Cantor decoder. The elementary loop invariants prove these effects for every natural command-store value under that boundary convention, not just for a bounded execution prefix.

Executable audit. The finite binding receipt is `composite_binding.json`. The command algebra supplement checks 164 symbolic cases for $n = 9$ and 188 for $n = 10$, with both opaque arbitrary tails and the exact zero-stack alias; an undecided equality is rejected. Separately, the independent whole-tape symbolic checker passes all 4,784 and 4,749 source outcomes and the exact 57,922-step blank startup. These checks support the displayed mathematical proofs and identify the literal implementations; they are not substitutes for an end-to-end kernel-checked arithmetic theorem.

7 Reduction experiments and what their failures establish

All searches in this section are finite. The grammar search found the 195-state construction above. Negative outcomes from other searches exclude only the explicit candidate families that their scripts enumerate. No optimality claim is made for 195. The earlier failures establish neither a minimum implementation cost for a logical presentation nor a general lower bound.

7.1 Parameter names and reassociation

The six unconditional heads allow independent renaming of their three parameters. For each of 101 good grammar shapes, every one of the $6^6 = 46656$ local permutation assignments was screened. This gives 4712256 assignments in total. None had optimistic effective AX reader cost below 62; on every shape the identity assignment was the only assignment reaching cost at most 63. Eighteen independent Python/C++ score comparisons checked the fast evaluator.

The word "optimistic" matters: reader matching is a lower estimate before all helper interactions are checked. The first joint renaming and reassociation runs reached apparent 61-reader cases that failed exact simultaneous-fold checks. Incorporating that compatibility check in the fast evaluator then found a valid 67-reader, six-fold grammar: an effective 61 readers and 195 total controls. The first successful run used only identity parameter permutations. Recorded results and seeds are in `construction195/search/`. This search tests new associations as well as fixed shapes; its exploration is heuristic rather than exhaustive.

7.2 Synthesized helpers and local state deletion

The new composites were tested against inherited good grammars, and one- and two-helper synthesis was tried. The helper templates add pure bit-preserving right-moving reader states and ask whether they enable enough extra compatible merges to pay for their own cost and save one net state. The optimistic filters found no 195-state candidate within the enumerated templates. Exact per-grammar counts and the source of the filters are included; no global decoder bound is inferred.

Independent context-based searches also test ordered control merges, one-helper repairs, independent changes to both retained rows, two-helper repairs, and predecessor-boundary bypasses. The recorded contexts overapproximate all relevant primitive frames, with joint register zero/nonzero information where used. Rejecting a candidate on an overapproximation is conservative for finding reductions: it can miss an invariant-dependent valid optimization. Surviving a representative context alone is insufficient, so all retained candidates are checked against the complete context family.

The two-helper screens consider more than 660 million patches for each composite when their first- and second-branch trials are combined. No candidate passed every required context. A positive control checks that the broader bypass checker accepts an independently constructed temporary-tape-tag simulation. This guards against mistaking a checker that rejects everything for useful negative evidence. Full counts,+assumptions, window widths and bounded step limits are in the portable checker and receipts.

7.3 Source shortening and wider macro boundaries

The source-level negation sequence `push;incT;cons` can be replaced by a 20-site arithmetic sequence in place of 47 sites, provided the required scratch register is zero at entry. Applying it at eight positions shortens the separate-head Combined program from 3649 to 3433 sites. The supplied extension audits the joint zero-mask invariant, the projected source CFG and the

complete runtime contracts. The best completed initial layouts have 197 controls, so this is a source improvement rather than the requested physical reduction.

A further search exposes closed factors inside push, cons and parameter copies, allowing the reader layout to share fragments across former macro boundaries. It keeps the primitive source CFG unchanged, checks actual compatible folds and validates any potential physical candidate. Its saved score and grammar record the bounded result. Again, physical control count, legal entry sides, path and scanner bounds, complete runtime simulation and blank startup all have to pass together.

7.4 Other inherited ideas tested or assessed

The 189-token joint-gate emitter and its 66-rule variable-arity dictionary were re-associated with the changed command fixed part. The tested binary layouts did not produce a valid sub-196 table. Some small reader counts had a nonzero feasibility penalty, so their objective values are not reported as working-state counts.

A direct transplant of ZFC K215's zero-valued appended spare policy was tested on the original and the two new PA tables. The tested starts failed to produce the required frame; the native 74 runs were still sweeping outward within the finite bound. This rejects the tested bootstrap transplant, not the zero-credit method in its own source machine. Its usefulness depends on a full initialization invariant.

The U208 history fold has a particular mixed incoming-direction motif absent from the published PA196 table. That observation excludes a literal relabelling of that one fold. It says nothing about more general context-dependent encodings or multi-step simulations.

ConciseIPRA supplies a valuable strategy of choosing a compact checker and transporting proof systems through explicit translations. It does not license replacing the searched calculus by iPRA without recovering the PA consistency calibration. Any such replacement must supply both directions at the level of PA-verifiable proof codes.

8 An exact direct-negation replacement

This section records a source simplification implemented by an experimental 197-state table. It does not meet the requirement of fewer than 196 states. The parent is the separated diagonal emitter with the Combined196 command control changes. Its source has 3,649 sites; the replacement has 3,433.

Write $T = r_1$, $W = r_2$, $S = r_6$, and $R = r_7$, and put

$$p(a, b) = \frac{(a + b)(a + b + 1)}{2} + b.$$

The old negation block is `push; increment T; cons`. The replacement is `drain T into S; increment T; pair-and-times-four`. It reuses the existing source subtrees `node37` and `node64`, whose native entry states are 28 and 106.

Arbitrary-value block identity. For every $t, w, s \in \mathbb{N}$, both blocks, entered with $(T, W, S, R) = (t, w, s, 0)$, terminate with

$$(T, W, S, R) = (4p(t + s, 1), w, 0, 0),$$

and preserve all other registers. In the old block, the push stores $p(t + s, w)$ in W and clears T, S, R. After the increment, $T = 1$. The cons block unpairs W back into $t + s$ and w and

computes $4p(t+s,1)$. In the new block, the first drain gives $T=0, S=t+s$; the increment gives $T=1$; the unchanged pairing suffix gives exactly the same result.

Here is an explicit termination argument for that pairing suffix, avoiding an inference from finite input tests. It first drains T into both S and R. At the following checkpoint $T=0$, and the invariant is

$$R+\frac{S(S+1)}{2}=p(s_0,t_0)$$

for the initial inputs $(S,T,R)=(s_0,t_0,0)$. A nonzero S is decremented once, R is incremented once, and the remaining S is drained into T. Returning through the first drain changes the checkpoint values by $(R,S)\mapsto(R+S,S-1)$, where the right-hand S is the old value. The displayed invariant is preserved, and S decreases strictly. Consequently this phase terminates with $S=T=0$ and $R=p(s_0,t_0)$. The final loop drains R while making four increments of T per iteration, giving $T=4p(s_0,t_0)$ and $R=0$. Every drain terminates by decreasing the register drained. The argument is an elementary induction over natural numbers and is formalizable in PA.

The entry condition $R=0$ matters. With all registers initially zero except $R=1$, the old 47-site block returns $T=16$, whereas the new 20-site block returns $T=12$. Thus this is not an unconditional replacement on arbitrary source configurations.

The actual source invariant. The clean seed is source site 25 with register vector $(2,0,0,0,0,0,0,0,0)$. A bit mask records, for each register, whether its value is zero or positive. Increment makes the corresponding bit positive; a decrement at zero takes only its zero branch; a decrement at a positive value takes its positive branch and nondeterministically leaves zero or a positive value. Other instructions preserve that bit. The computed overapproximations contain 263,244 old and 247,308 new site/mask pairs. The replay independently checks initial inclusion and closure under every abstract successor. Induction on a concrete execution therefore places every reachable configuration in the abstraction. At each of the eight replaced entries, every admitted mask has bit 7 clear. Hence the required $R=0$ hypothesis holds without sampling a run.

The old entry ordinals are

$$584, 686, 1759, 1861, 3007, 3081, 3316, 3472;$$

the new entries are

$$584, 659, 1705, 1780, 2899, 2946, 3154, 3283.$$

The regions are disjoint. Their only external entry is their first site and their only external exit their following site. The 3,273 other source rows agree under the order-preserving relocation, including all branch targets. The local identity and entry invariant therefore compose into a halting-preserving source simulation from the clean seed. This argument compares source computations; it does not substitute a finite blank-tape prefix for a proof of nonhalting.

Literal implementation and limits. The table has 197 working states, with 68 axiom-reader states and five compatible reader folds over the fixed 134-state part. The complete symbolic primitive-runtime audit has 4,461 cases and a maximum of 497 symbolic steps. The literal startup reaches the exact source frame after 57,922 steps. The checker uses the unchanged upstream arbitrary-register block contracts; these executable certificates are not newly exported Lean theorems. The mathematical and encoded-proof calibration of the parent calculus remains a separate dependency. No state saving or new end-to-end PA biconditional is claimed for this 197-state experiment.

9 The proof endpoint and remaining obligations

The imported PA196 formal project explicitly exports

$$\mathsf{PA} \vdash \mathrm{Nonhalt}(\mathrm{PA196}) \to \mathrm{Con}(\mathsf{PA}).$$

Its documentation also records external nonhalting and the corresponding independence results. It does not list the reverse implication inside PA as an exported theorem. This is an inventory statement about the formal endpoint, not a claim that the converse is false or that an ordinary mathematical proof must be rejected for lacking Lean code.

The incoming manuscripts supply substantial ordinary mathematical arguments for the converse. They interpret membership by arithmetic binary-bit membership, equality by arithmetic equality, and universal coding with the stated transposition. A PA proof template using full induction generates the finite comprehension instances needed for the separation gate. The live-store argument must track derivations for the top, cached head and finite stack, preserving the code 0 truth alias and handling coincident numerical labels. A physical halt must then yield a source halt and a finite contradiction proof.

The new command proof establishes its own part of this chain. The literal table contracts establish another part relative to the inherited block interfaces. An end-to-end kernel-checked certificate would additionally bind the concrete arithmetic proof-code transformations, the chosen numerical calculus, the scheduler, and the extraction of a source trace from every physical halting trace, in the intended arithmetization. The repository's standard-natural-number certificate stack developments are useful prerequisites; they should not silently be promoted into an additional formal endpoint without that work.

For the primary construction, these inherited logical interfaces are exactly the parent's interfaces: the calculus, command program and resolved source instructions are unchanged. The new work supplies a different physical implementation and the two-way preservation proof. The mathematical 195-state consistency result therefore composes with the parent's stated calibration. The reproducible finite certificates are evidence for the new implementation, and are not described as an independent Lean verification of every inherited theorem.

This distinction does not change the literal state count: the primary table has 195 working controls under the same convention as its 196-state parent. It explains precisely which proof is new, which calibration is reused, and what the executable audits certify.

10 Reproducing the results

The archive is self-contained for the new 195-state table reconstruction, runtime/source audits and the secondary command audits. It includes Python source, C++ search source, native and dense tables, grammars, source paths, receipts, source licenses and provenance. Python 3.10 or newer and a C++17 compiler are sufficient for these portable checks; the LaTeX document requires a normal pdfLaTeX setup. The large upstream Lean development is identified by revision rather than copied wholesale.

From the root of the extracted archive:

```
python3 construction195/build_check.py
python3 construction195/build_check.py --combined
python3 construction195/rotation_witness.py
python3 source195_audit/audit.py
python3 physical195_audit/replay.py
python3 axioms/verify.py
```

```
python3 independent_verification/check_composite_commands.py
python3 independent_verification/check_composite_binding.py
python3 independent_verification/replay.py
```

The default replay uses recorded context certificates. Use `--full` to derive the complete symbolic source contexts again; `--extended` also replays the two-helper screens. These optional search replays may take substantially longer than the direct table audit. The exact commands for the parameter and negation extensions are in their accompanying README files.

To reproduce the full source census, obtain the repository at the pinned revision and run the inventory scripts using the commands in `inventory/COMPARISON.md`. Those scripts inspect archive members and table bytes; they do not execute code found inside the reports. The release manifest lists SHA-256 hashes for every delivered file except the manifest itself. A manifest validates byte identity, not mathematical correctness.

To compile the report:

```
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

11 Sources and attribution

The work reuses source and mathematical arguments from James Chang's PA196/PA209 projects and the reports collected by Vladimir Reshetnikov. The included source excerpts retain their relevant licenses and provenance. New tables are explicitly distinguished from the source tables. References below point to the fixed repository revision.

1. ProveIt: PA research directory, including `jamesyc-upstream/README.md`, the imported `pa196/docs/result.md` and `docs/reproduction.md`, and the comparison/PA-bridge documents.
2. ProveIt: all 27 incoming archives. The appendix identifies each filename, and the census records each member and hash.
3. `pa_consistency_comparison_and_new196_2026-11-13.zip`: diagonal and separate-head grammars, formula-emitter audit and simultaneous-fold compiler used as the first composition parent.
4. `conpa_tm_study_2026-11-13.zip`: Combined196 commands, physical row edits, command algebra and source-binding methodology used as the second parent.
5. `pa_tm_research_2026-11-13(2).zip`: source 26 CD6 specialization, joint zero-mask abstraction, scanner and parity experiments, used for cross-checks and failed-candidate comparison.
6. ProveIt: ZFC research directory, consulted for physical reader, spare-credit and bootstrap ideas.
7. ProveIt: ConciseIPRA directory, consulted for compact checker and proof-translation design principles.

A Comparison of all supplied PA machine designs

A.1 Scope and principal finding

The source snapshot is ProveIt commit `906367250a7f96c923147b10df392294bced7be5`. The census reads every `.tm` file in `Computability/BusyBeaver/Research/PA` and every `.tm` member of every incoming ZIP, following nested ZIPs and scanning each byte-distinct ZIP payload once. It executes no submitted code. It separately checks the designated initial controls and halt controls for the curated final tables. The complete census includes ancestors, rejected experiments, raw intermediate compilations, and synthetic controls; its broad counts must not be advertised as counts of valid consistency machines.

The supplied source corpus contains **no accepted sub-196 construction**. There are six root-free graph classes of explicit 195-state diagnostics, all rejected or unproved by the source reports. Two separate rejected 196-state layouts actually halt on the blank tape; those finite halts were independently replayed here. There are more positive same-size designs than the older repository catalogue records: the updated explicitly started catalogue has 23 distinct 196-state graphs, comprising 19 baseline or report-positive designs, two explicitly experimental MP designs, and two blank-halting controls.

This comparison is not a new proof of the PA equivalence of any table. The term **report-positive** records the source manuscript's claim of a successful construction, with the stated executable checks. The exact proof scope is kept separate below.

The new 195-state constructions obtained during the present work are outside this pinned-source census and are described separately. Their existence does not change the historical counts or turn the six supplied diagnostics into accepted constructions.

A.2 Freshly checked counts

Quantity	Count
Top-level incoming ZIP files	27
Byte-distinct ZIP payloads, including nested historical inputs	40
ZIP references encountered, including repeated payloads	68
Maximum ZIP nesting depth	8
Transition-table occurrences in distinct ZIP payloads	1,466
Additional `.tm` files in the PA tree	14
Total table occurrences	1,480
Byte-distinct table encodings	531
Distinct parsed tables retaining literal state labels	522
Malformed or incomplete binary tables	0
Canonical root-free action-graph classes with a vertex reaching all controls	401
Additional synthetic six-state control without such a vertex	1
Designated 196-state tables / rooted graph classes	23 / 23
Designated U208 presentations / rooted graph classes	27 / 20
Explicit 195-state file occurrences / literal encodings / root-free classes	14 / 8 / 6

There are therefore 402 root-free action-graph classes in the broad corpus: 401 obtained by canonical traversal plus the unique six-state synthetic control, which cannot be isomorphic to any other table because no other table has six states. Among 196-state tables the broad corpus has 90 root-free graph classes; most are exploratory tables, not designated finals. Root-free classification deliberately ignores the initial control and must not be used as a claim of equal

blank-tape behavior. The 23-table designated classification preserves the initial control, read bit, write bit, direction, and halt control, and checks graph reachability of all 196 working controls.

Detailed outputs are `census/tables.json`, `tables.csv`, `graph_classes.json`, `graph_classes.csv`, `rejected195_graphs.json`, and the two `designated*` directories. The parse and graph algorithms are in `inventory.py`, `graph_census.py`, and `designated_census.py`.

A.3 Main lineages

Machine family	Working states	Source sites	Distinct internal source nodes	Maximum source path	Reported blank startup
James Chang PA209	209	3,948	173	33	58,210
U208 A	208	3,909	174	42	48,140
U208 B	208	3,909	174	41	48,140
Published PA196	196	3,812	159	38	58,154
Shorter-path alternative196	196	3,812	159	36	58,154
Combined196: H-only Shift plus MP result in H	196	3,788	—	—	57,922
Diagonal-vacuity196	196	3,700	—	—	58,154
Separate diagonal-head 196	196	3,673	—	—	58,154
Separate-head scanner-oriented196	196	3,673	—	—	58,294
Source-26 CD6 zero-jump196	196	3,812	—	—	58,402

The source and startup numbers in this table are manuscript and receipt claims, not newly recomputed by this inventory worker. Table structure, identity, and graph classes were freshly recomputed. The two invalid blank halts described below were also freshly replayed.

James Chang's imported snapshots are PA196 commit `407cdc356f6b586635e71ac92be394255f4bfea4` and PA209 commit `69d98241a721ead20b4f0b17df9062950b40fbfe`. The U209 parent of the U208 lineage is a different source program from James Chang's PA209. Native control labels have local meanings; matching state numbers across these lineages is not a simulation argument.

PA196's published budget is 36 arithmetic-engine controls, 13 leaf routers, and 147 reader controls representing 159 distinct internal source nodes. Its six complementary AX reader folds are already included in 196. The PA209-to-PA196 logical changes include the transposed universal code `U(a,B)=4pair(B,a)+2`, removal of the old seventh head, guarded equality vacuity, altered head and antecedent orders, and complementary reader sharing.

U208's last fold identifies native 211 with 172 using incoming-history information and a helper. Its parent contains the mixed-entry motif `163,0->0L211` and `163,1->1R211`. Published PA196 contains no control with the corresponding pair of actions to a common working target. That excludes an exact relabelling of this particular fold, not other contextual or offset simulations.

The U208 outer interpreter also cyclically moves the first 25 source sites to the end and uses a simpler Shift accepting on the head alone. Combining that outer interpreter with PA196 AX has a 196-state runtime prototype in the newest archive. Its 4,957 arbitrary-value source-frame contracts pass, but every existing initial control follows a blank startup ending in an outward loop. The attempted 197-state prefix repair breaks a runtime case. It is therefore not a complete blank-start consistency machine.

A.4 All 27 incoming archives

Every filename below is present at the pinned revision. Hashes, direct table counts, and exact member paths are in the machine-readable inventory.

Archive	Main contribution	Result or limitation
`Below_U208_Research_2026-10-29.zip`	Source-boundary windows, offset quotients, guarded source pruning	3,909 to 3,904 source sites; no validated machine at or below 207
`ConPA_below208_research_2026-10-29.zip`	Comparative audit, finite-model obstruction to cover-role identification, local folds	No reduction below 208; template-specific failures are not global lower bounds
`PA_consistency_TM_research_2026-11-13.zip`	Exact original-source grammar with 159 internal nodes and zero-run bound five; exact 157-node combined-negation/MP grammar; residual MP	No validated sub-196 table or valid shorter-scanner startup
`U208_ConPA_2026-10-29.zip`	Three-cell history fold of native 172/211	One presentation of original U208 A
`U208_ConPA_Conditional_Context_2026-10-29.zip`	Conditional-entry proof, primary and alternative tables	Original U208 A/B, not additional independent state savings
`U208_ConPA_Entry_Context_2026-10-29.zip`	Complete entry-context proof and tables	Same original A/B classes
`U208_ConPA_History_2026-10-29.zip`	Finite-history presentation and local stuttering proof	Same original A/B classes
`conpa_tm_comparison_and_reduction_search.zip`	Two experimental MP196 variants; layout and register catalogues	Two proposed Shift layouts halt on blank after 43,563 steps
`conpa_tm_comparison_and_search.zip`	A/D register-swap196; two-rotation195 quotient; logical countermodel	Quotient fails 93 source contracts
`conpa_tm_study_2026-11-13.zip`	Combined196 (H-only Shift and MP result in H); 3,788 sites, 4,928 contracts; shorter-negation197	Complete same-size intermediate; no strict reduction
`pa196_research_2026-11-13.zip`	Exact 157-node representation preserving original source CFG; HA-strength reference calculus	Reader-context savings are lost in physical allocation
`pa208_research_2026-10-29.zip`	Eighteen additional U208 helper variants; source and theory work	Enlarges U208 to 20 designated graph classes, none below 208
`pa_below196_research.zip`	Updated earlier census; eight equivalent separation gates; exact restricted Boolean formula-size result; grammar and physical screens	No validated sub-196 construction; includes prior failed195 tables as diagnostics
`pa_consistency_comparison_2026_11_13.zip`	Modal-B plus explosion split; finite Hilbert proofs; twelve composition experiments; 189-token joint gate	Physical constructions remain 196–200 states
`pa_consistency_comparison_and_new196_2026-11-13.zip`	New diagonal196 and separate-head 196; scanner-oriented196; stronger simultaneous-fold checks; runtime-valid hybrid	Three new alternate-calculus196 tables and one original-calculus scanner196; naive195 scanner fails
`pa_consistency_reduction.zip`	Rebracketed196, H-only196, integrated-witness gate, internal-converse manuscript	Gate compiles to 203/206; failed195 helper combinations

Continued on next page

Archive	Main contribution	Result or limitation
pa_consistency_research.zip	Exact 158-node source grammars and scanner-repaired196 layout	Nominal195 mistakes an address run for the marker; repair costs one state
pa_consistency_research_2026-10-29.zip	Inductively certified 3,909-to-3,904 source specialization	No validated physical machine below 208
pa_consistency_research_2026-11-13(2).zip	Twelve-rotation196; diagonal implementation199-to-198; successful-MP halt; joint register abstraction	No strict improvement below 196
pa_consistency_research_2026-11-13.zip	Diagonal-vacuity calculus and 3,700-site199 implementation; short-negation197	Logical simplification does not save physical states in that layout
pa_tm_comparison_and_reductions.zip	Rotated196, macro62-rotated196, distributed gate with 3,773 sites	Distributed gate compiles to 198; grammar counts are not physical counts
pa_tm_extended_research_2026-11-13.zip	Four closing-CD3/CD4/CD5/CD6 variants, stronger source invariants	All196 and all change the same native row, so their changes cannot be stacked
pa_tm_research_2026-11-13(2).zip	New source-26 CD6 zero-jump196; complete zero-mask invariant; rejected195 repair search; integrated base-eight parity runtime	New same-size table; base-eight202 runtime has no certified sound blank seed
pa_tm_research_2026-11-13.zip	Shorter-path alternative196; symbolic checks; physical Lean subset; direct PA proof-translation manuscript	Same-size candidate; alternative-specific full formal endpoint not exported
pa_unary_power_cover.zip	Unary weak-covering theory with mutual PA interpretation and proof templates	No newly compiled literal machine
u208_comparison_research_2026-10-29.zip	Original two-class U208 census, weak/exact schema comparison, rejected207 frame witness	Predates eighteen helper classes; no sub208 result
u208_research_2026-10-29.zip	Two U208 outputs and symbolic runtime evidence	Same original history-fold saving

A.5 The 23 designated 196-state graphs

Every row below has 196 graph-reachable working controls and 392 transitions. These are 23 distinct rooted action graphs, not merely different filenames. The full hashes and source member paths are in census/designated196/catalogue.json.

Design	SHA-256 prefix	Status and distinction
Published PA196	e8bbccd4cfa3	Upstream formal development
MP result in H	893f08672746	Explicitly experimental command change
MP without zero push	3d7f91e548f8	Explicitly experimental command change
Rejected Shift layout	db5cd7df1475	Blank halt at 43,563
Rejected combined layout	d3a049def261	Blank halt at 43,563
Shorter-path alternative196	aa05f079326a	Head order and association; maximum path 36
Closing CD3	d5c1c2c69082	Forced-zero decrement in place of closing jump

Continued on next page

Design	SHA-256 prefix	Status and distinction
Closing CD4	`2be1986cbbac`	Same row, different zero register
Closing CD5	`4433af83b568`	Same row, different zero register
Closing CD6	`2db09bb05768`	Same row, different zero register
Rotated196	`25d686ab95f5`	Source-CFG-preserving rotations
Macro62 rotated196	`591f42407752`	Alternative rotated reader layout
Regrouped158-node196	`1ab0fd1fe7b7`	Shorter source grammar plus longer scanner
A/D register-swap196	`3968e6733b30`	Physical register permutation
Rebracketed196	`e212e359837c`	Different source-tree association
H-only Shift196	`5e7e8daf0fbd`	3,811 source sites
Twelve-rotation196	`b9873b89df41`	Later source-CFG-preserving layout
Combined196	`bd74754d75e7`	H-only Shift plus MP-result-in-H; 3,788 sites
Diagonal-vacuity196	`59bbe3dbe040`	191 AX macro items; 3,700 sites
Separate diagonal-head 196	`61ffdfea87d2`	190 AX macro items; 3,673 sites
Separate-head scanner196	`bccb07e7da72`	69 AX readers and seven mutually compatible folds
Original-calculus scanner15	`ed5454b2db8b`	Fifteen remaining address obstructions; native start 74, halt 254
Source-26 CD6 zero-jump196	`2c1a736399e0`	One native row27/1 becomes `1R21`

The four closing-decrement variants all modify native `(1365,1)`, dense `(193,1)`. Their individual row choices are alternatives. The new source-26 zero-jump changes native `(27,1)` instead, so it is a different source specialization; composing it with a closing specialization still needs whole-program runtime checks because helper paths can interact.

The separate-head calculus pushes ten rather than nine items in AX. The source compiler's head-selection Shift counts must therefore change. Simply swapping its emitter into the old proof-command compiler would leave incorrect selection constants.

A.6 Every explicit 195-state diagnostic

Graph family	Encodings	Supplied rejection evidence
Two-rotation quotient	Native and dense	93 of 4,959 source contracts fail; an ideal unimplemented history router is insufficient as a literal machine
`warm_physical_61_8047`	One	714 failed contracts; a valid source frame at site 1111 enters an exact four-transition cycle after 26 steps
`extended_physical_61_12085`	One	777 failed source contracts recorded by its supplied receipt
Seven-zero regrouped scanner	One	Site1003 is confused with the marker; wrong clean frame after 276 steps; scanner repair returns 196
Separate-head short-scanner	Native and dense	96 unrestricted failures; even the intended clean seed reaches a root after 118 steps with 43 wrong tape cells
Wrong-opcode zero-jump fold	One sparse dense-labelled table	Native27/1 is incorrectly redirected to `1R62`; source 26 reaches wrong site 27 with register 6 incremented and an extra spare, instead of site 34

These six graph families appear in 14 file occurrences because reports repackage earlier diagnostics. Their eight literal encodings are not eight different constructions. Most failures refute a claimed

source simulation, not every possible account of the table's eventual blank behavior. The wrong-opcode parent also exhibits that wrong source operation on its actual blank run. Its 195-state child cannot be validated just because a local fold is correct relative to the already incorrect parent.

A separate code-defined dense45-to160 merge with repair184/1=1L75 survives100 million blank steps yet fails its claimed simulation at the parent boundary 1,223,335,721, selecting source 615 rather than 605. It is recorded in the integrated PA search material; it is not an extra explicit .tm entry in the six-family count. This is a useful warning about long finite-prefix evidence.

The two rejected196 controls have a stronger and independently reproduced failure: both halt after exactly 43,563 transitions on blank, in halt 196 with head 1 and the same complete finite-support tape. Their sole difference at row 115/1 is not visited by that run. See census/designated196/rejected_blank_halt_replay.json.

A.7 Proof boundaries

The precise imported formal PA196 result is

$\mathtt{PA} \vdash \mathtt{Nonhalt(PA196)} \rightarrow \mathtt{Con(PA)}$.

The imported Lean project also establishes external nonhalting in its metatheory. Its theorem inventory does not export the reverse implication inside PA. The same distinction applies to imported PA209. External soundness, independence in both directions, and a finite simulation prefix do not supply the reverse implication by themselves.

Incoming manuscripts contain substantial ordinary mathematical arguments for the converse: translate membership to binary-bit membership, equality to arithmetic equality, and the transposed universal constructor directly; generate PA proofs of finite comprehension by substituting the input formula into a full-induction proof template; preserve code0's truth alias; handle coincident source labels; and carry proofs through the live store and stack. Their positive-length exact macro simulations plus determinism are intended to extract a source refutation from any physical halt. Absence of a new Lean certificate does not alone refute these mathematical arguments. Conversely, the existence of the manuscripts does not mean every exact proof-code and machine-binding obligation has already been checked.

The integrated PA directory includes useful standard-Nat prerequisites: canonical certificate-stack treatment of pair(0,0)=0, bindings to actual command definitions, a numeric native checker, and checker-rule audits. The documentation keeps the arbitrary-PA-model trace extraction and the Foundation model-valued proof-code bindings as separate obligations. A replacement table must preserve the appropriate command compiler, complete runtime contracts, and exact blank startup as well as the logical theory.

ConciseIPRA supplies proof-checker design principles. Consistency of iPRA cannot simply replace consistency of PA; an encoding through an intuitionistic checker requires the full PA-strength calibration. ZFC K215 supplies useful physical techniques, but its numerical control labels and code conventions do not identify states in the PA196 table.

A.8 Opportunities identified in the supplied designs

1. The separate-head 190-item emitter has substantially fewer source sites while still costing196 controls. Its final all-helper compatibility checker is a stronger search base than a matching-size estimate alone. The earlier optimistic70-reader/nine-fold195 count failed because of invalid macro entry sides and interacting helper paths. This direction led to the new construction: an unchanged 190-token macro word is reassociated from68 distinct readers to67, retaining six valid simultaneous folds and giving134+67-6=195 working controls.

2. The scanner-oriented separate-head layout has only six static internal address obstruction-s:197,584,786,867,1759,2068. Removing these is not by itself a correct scanner deletion; dynamic address bits, reader aliases, and startup require a joint invariant.
3. The runtime-valid U208/PA196 hybrid isolates blank initialization as a concrete unsolved issue. Its initial-control screens and unused-row census are supplied. Any successful bootstrap must also preserve the universally quantified runtime cases.
4. Source-26 zero-jump, closing-jump specialization, H-only Shift, MP-result-in-H, and the diagonal/separate-head calculus involve different source regions but share physical helper paths. Their composition is a promising question only after full simultaneous verification; savings cannot be summed arithmetically.

No negative result cited here proves a global lower bound of 196.

B Inherited diagonal-calculus calibration

This appendix reproduces the diagonal-interface argument from the supplied diagonal-calculus report, as carried in the parent archive. Its historical physical implementation paragraph is omitted; the195-state construction in this report preserves the later separated-head implementation. Replace the equality-vacuity component of the gate by the diagonal clause

$$\Gamma(B) = I(B, U(B, B)), \qquad \text{that is,} \qquad B \longrightarrow \forall x_B\, B. \tag{1}$$

Here B is both a formula code and a variable index. This clause is sound because every variable index in the decoded formula B is strictly below B. Code zero is truth and presents no exceptional free variable.

For the implementation included in this archive, replace the old gate

$$F(a, b, c) = \neg(\Delta_c(a, b) \to (V(a, b, c) \to Q(a, b, c)))$$

by

$$F_\Gamma(a, b, c) = \neg(\Gamma(c) \to (V(a, b, c) \to Q(a, b, c))). \tag{2}$$

Keep the old guard $c \neq a, b$, the six other heads, the restricted propositional head, generalization, and modus ponens. Thus the membership-vacuity part of the gate is retained. Removing that part as well would invalidate the reverse-rewriting argument below.

B.1 What can be extracted from the new gate

Classical implicational reasoning gives both

$$F_\Gamma(a, b, c) \to \Gamma(c), \qquad F_\Gamma(a, b, c) \to G(a, b, c), \quad G = \neg(V \to Q).$$

Every diagonal instance $\Gamma(B)$ is derivable: take $c = B$ and $a = b = B + 1$. Both guard comparisons succeed. The extracted old gate G continues to give unrestricted membership vacuity at labels different from the atom's indices, and the original pointed-separation schema. Coincident labels $a = b$ are permitted by the guard and have already been accounted for in the arithmetic interpretation.

The retained propositional and quantifier heads give the same basic implicational reasoning, universal monotonicity, and the existential-elimination rule for a conclusion known to be vacuous at the quantified label. Membership vacuity extends to the quantifier-free membership matrices used below by the original negation and implication arguments. It also extends through a universal quantifier whose binder differs from the new vacuity label, by quantifier commutation. No equality-vacuity premise is used in those extensions.

B.2 Recovering reverse rewriting

The old proof used equality vacuity to discharge an existential witness from a conclusion containing an equality atom. The replacement is to make the separator property the negation of that entire conclusion.

Let $a \neq b$. Let $\chi(t)$ be one of $M(d,t)$, $M(t,d)$, or $M(t,t)$, with $d \neq a$ in the first two cases. The direct membership-rewriting heads give

$$E(v,y) \to (\chi(v) \to \chi(y)).$$

Put

$$C = E(a,b) \to (\chi(b) \to \chi(a)), \qquad \varphi = \neg C, \qquad c = \ulcorner \varphi \urcorner.$$

The code c exceeds all variable labels in C, so $c \neq a, b$; in the first two families it also gives $c \neq d$. There is no property-code collision requiring a separate padding branch.

Use pointed separation at (a, a, c). It gives

$$\exists x_c\,(S \wedge P), \qquad P = \varphi \to M(c,a), \qquad S \to U(a, M(c,a) \to \varphi). \tag{3}$$

Propositionally, φ implies $E(a,b)$, $\chi(b)$, and $\neg\chi(a)$. In particular, weakening the last consequent in (3) gives

$$S \to U(a, M(c,a) \to \neg\chi(a)).$$

The original negative-property specialization lemma now applies. It uses the equality witness head, the direct membership rewrites, and membership-only vacuity, and yields

$$S \to (M(c,b) \to \neg\chi(b)).$$

The equality atom occurring inside φ is never substituted: it was removed from the quantified consequent by the propositional weakening step.

Under φ, the point condition P gives $M(c,a)$. Direct rewriting under $E(a,b)$ gives $M(c,b)$, and the displayed consequence of S contradicts $\chi(b)$. Therefore

$$\varphi \to (S \to \neg P)$$

is derivable. Generalize at c, use universal distribution, and apply the available diagonal axiom $\Gamma(\varphi) = \varphi \to U(c,\varphi)$. This gives

$$\varphi \to U(c, S \to \neg P) = \varphi \to Q.$$

Pointed separation gives $\neg Q$. Hence $\neg\varphi$, and classical double-negation elimination gives C. This proof avoids a circular appeal to equality vacuity for C.

For $a = b$, the desired reverse rewrite is propositional. The cases $d = a$ reduce, as in the original proof, to direct rewrites and the already established diagonal case $\chi(t) = M(t,t)$. Thus for all labels the two unrestricted membership rewrites in both directions are recovered:

$$\begin{aligned} &E(a,b) \to (M(d,a) \leftrightarrow M(d,b)),\\ &E(a,b) \to (M(a,d) \leftrightarrow M(b,d)). \end{aligned} \tag{4}$$

B.3 Recovering the neighborhood-equality interface

Define

$$R(a,b,z) = U(z, M(a,z) \leftrightarrow M(b,z)) \wedge U(z, M(z,a) \leftrightarrow M(z,b)).$$

Let $z_0 = \ulcorner E(a,b) \urcorner$. Then $z_0 > a, b$, and the diagonal clause for $E(a,b)$ gives

$$E(a,b) \to U(z_0, E(a,b)).$$

Lifting (4) at z_0 gives $E(a,b) \to R(a,b,z_0)$.

It remains to recover the original interface at *every* $z \neq a, b$, not just at a numerically large binder. If $z = z_0$, nothing further is required. Otherwise consider either quantifier-free membership matrix

$$P(t) = M(a,t) \leftrightarrow M(b,t) \quad \text{or} \quad P(t) = M(t,a) \leftrightarrow M(t,b).$$

The two-direction rewrites imply $E(z,z_0) \to (P(z_0) \to P(z))$. The witness $\exists x_{z_0} E(z,z_0)$, universal distribution, and membership-only vacuity give

$$U(z_0, P(z_0)) \to P(z).$$

The antecedent is itself membership-only and vacuous at z, so generalization and distribution yield

$$U(z_0, P(z_0)) \to U(z, P(z)).$$

Apply this to both matrices. Consequently

$$E(a,b) \to R(a,b,z) \qquad (z \neq a, b)$$

is derivable without unrestricted vacuity for primitive equality.

The equality-witness head is unchanged. Existential monotonicity and the recovered neighborhood implication supply its interpreted witness form. These are exactly the original completeness proof's three interfaces: two-direction membership rewriting, neighborhood equality, and an equality witness, conventionally denoted (R), (N), and (W).

B.4 PA calibration and logical scope

The original PA-refutation compiler uses primitive equality through those interfaces, together with the old gate and the unchanged heads. In the inspected logical and arithmetic-certificate source closure, the uses of equality vacuity occur in the reverse-rewriting and neighborhood modules. Substitute the finite derivations above for those affected templates. Code formation, choosing $c = \ulcorner \neg C \urcorner$, and $z_0 = \ulcorner E(a,b) \urcorner$, the label-equality case splits, and the finite proof concatenations are primitive recursive. Their correctness is provable in PA. This gives the internal refutation-to-halting direction for the revised calculus.

In the other direction, each new diagonal clause has a uniformly generated first-order proof under the arithmetic interpretation, by the code-freshness lemma. Pointed separation is still supplied by full PA induction as in the inherited arithmetic-calibration argument. The annotated proof-store argument and positive-length tape simulation then give the internal halting-to-refutation direction.

Thus the written calibration argument establishes the same target:

$$\mathsf{PA} \vdash (\neg H_\Gamma \leftrightarrow \mathrm{Con}(\mathsf{PA}))$$

for a correct literal implementation of the revised program. This is a calibration to PA; merely replacing the searched system by iPRA would not supply it. The construction applies the general strategy of replacing a primitive schema by smaller derivably equivalent interfaces. Its decisive freshness lemma comes from PA196's existing numeric coding: the magnitude of a formula code guarantees a fresh variable label. We do not attribute that specific diagonal trick to the ConciseIPRA proposals.

C Complete primary transition table

This is the complete dense table of M_{195S}. The initial state is 0, the blank symbol is 0, and the unique terminal label is 195. The head starts at cell 0 of a two-way infinite zero tape. Each entry wDq means write w, move $D \in \{L, R\}$, then enter q. Columns 0 and 1 are the read symbol. There are 195 working controls and 390 transitions; the halt label has no rows.

q	0	1	q	0	1	q	0	1	q	0	1
0	1R1	1R2	49	0R69	0L14	98	0R87	1L115	147	0R134	1R157
1	1R3	1R1	50	0R70	0R71	99	0R104	0L14	148	0R158	1R159
2	0R0	1R2	51	0R72	0L49	100	0R36	1R116	149	0R160	1R160
3	0R4	0R1	52	0R73	1R54	101	0R115	1R78	150	0R138	1R161
4	1R5	1L6	53	0L74	0L15	102	0L46	0R113	151	0R147	1R162
5	1L7	1L8	54	0R75	0L10	103	0L45	1R117	152	0R163	1R164
6	0L7	1L6	55	0R76	1R77	104	0L94	1R118	153	0R134	1R165
7	0L9	1L6	56	0R78	1R79	105	0R119	0L14	154	0R120	1R166
8	0L10	0L11	57	0R80	1R81	106	0R71	1R106	155	0R138	1L143
9	1R12	0L9	58	0R82	1R83	107	0R120	1R30	156	0R154	1R167
10	0L9	0L10	59	0L84	0R1	108	0L46	0R106	157	0R162	1R168
11	1L8	1L11	60	0R85	1R60	109	0R70	0R95	158	0R169	1R170
12	0R12	0R13	61	0L76	1L86	110	0R59	0R95	159	0R171	1R164
13	0L14	0R13	62	0R87	0L7	111	0L65	1R121	160	0L172	1R132
14	1L15	0L15	63	0R88	1R89	112	0R122	1R123	161	0R173	1R174
15	0L15	1L16	64	0R42	1R90	113	0R14	1R113	162	0L175	1R41
16	0L17	1L16	65	0R91	1R49	114	0R68	1R114	163	0L137	1R175
17	0L18	1L16	66	0R92	0L75	115	0R124	1L67	164	0R176	1R171
18	0L19	1L16	67	0R86	1R33	116	0R125	1R126	165	0R138	1R142
19	0L20	1L16	68	0R93	1R68	117	0R120	0L14	166	0R177	1R120
20	0L21	1L16	69	0L94	1L33	118	0R94	1L98	167	0R178	1R179
21	0L22	1L16	70	0L84	0R2	119	0R127	1R127	168	0R180	1R181
22	0L23	1L16	71	0R95	1R71	120	0R70	0R68	169	0R120	1R163
23	0R23	1R24	72	0R75	1R96	121	0R128	1R129	170	0R182	1R155
24	0R25	1R26	73	0R59	0R85	122	0R32	1R36	171	0R134	1R139
25	0R27	1R28	74	0R86	1R97	123	0R31	1R79	172	0R155	1R183
26	0R29	1R30	75	0R42	0L13	124	0R59	0R114	173	0R184	1R178
27	0R31	1R195	76	0R87	1R37	125	0R102	1R117	174	0R160	1R185
28	0L32	1R33	77	0R30	1R98	126	0R82	1R81	175	0R135	1R130
29	0R34	1R35	78	0R42	1R99	127	0R120	1R120	176	0R165	1R41
30	0L36	1L37	79	0R100	1R101	128	0R130	1R131	177	0R132	1R183
31	0R38	1R39	80	0R102	1L103	129	0R132	1R133	178	0R179	1R135
32	0R40	1R41	81	0R30	1R80	130	0R134	1R135	179	0R139	1R41
33	0R42	1R43	82	0R97	1R104	131	0R136	1R137	180	0R32	1R186
34	0R44	1R45	83	0R30	1R105	132	0R138	1R139	181	0R172	1R167
35	0R46	1R47	84	1R84	1L15	133	0R140	1R141	182	0R187	1R138
36	0R48	1R49	85	0R106	1R85	134	0R100	1R67	183	0R187	1R130
37	0R50	1R51	86	0R73	1R107	135	0R79	1R142	184	0R41	1R187
38	0R52	1R42	87	0L46	0R93	136	0R28	1R143	185	0R184	1R188
39	0R53	1R54	88	0R108	1R98	137	0R144	1R145	186	0R79	1R162
40	0R55	1R56	89	0R108	1R69	138	0R100	1R146	187	0L173	1R41
41	0R57	1R58	90	0R109	0L14	139	0L147	1R120	188	0R176	1R189
42	0R59	0R60	91	0R110	1L74	140	0R148	1R149	189	0R190	1R191
43	0R61	0L14	92	0R111	1R112	141	0R150	0L10	190	0R179	1R91
44	0R62	1R63	93	0R113	1R93	142	0R41	1R127	191	0R192	1R193
45	0R55	1R64	94	0R50	1R103	143	0R151	1R152	192	0R31	1R53
46	0L65	1R66	95	0R114	1R95	144	0R153	1R154	193	0R194	1R192
47	0R36	1R67	96	0R30	1R49	145	0R155	1R156	194	0R96	1R115
48	0R59	0R68	97	0R48	1R61	146	0R91	1R64			

Dense-table SHA-256: `30246dfe308e4b406e3bbf461f511530d49112974f90972327cf950ba41f919f`.